17-SEP-2022

Inferential Statistics



→ In order to validate, we use hypothesis testing.

Hypothesis Testing

Steps of Hypothesis testing

① Null hypothesis

- Default statement will be null hypothesis. By default null hypothesis is always true.

e.g. A person has commited crime or not.
Null hypothesis ÷ A person has not commited crime.

e.g. whether a coin fair or not.
$P(H) = 0.5$ and $P(H) = 0.5$ ∴ coin is fair
Null Hypothesis : coin is fair.

② Alternate hypothesis

- Always be the opposite to the null hypothesis.

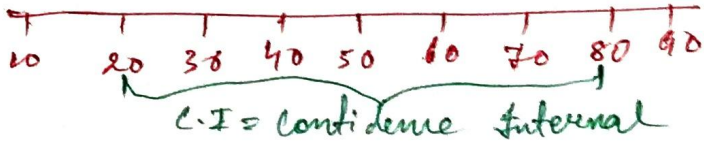
e.g. whehther a coin is fair or not.
Alternate hypothesis: coin is not fair.

e.g: A person has commited crime or not.
Alternate hypothesis : A person has commited crime.

③ Perform Experiment

- In order to check coin is fair or not,
- A coin has been tossed 100 times.

Head = 50 times — Tail = 50 times } Fair
Head = 60 times — Tail = 40 times
Head = 70 times — Tail = 30 times } not fair based on confidence interval
Head = 80 times — Tail = 20 times



→ A domain expert will tell the range (confidence interval) to check the coin. If the no. of times head will fall b/w range given by domain expert, then we will say that coin is fair. And we will accept the null hypothesis.

→ Head = 60 times (lies b/w C.I)
Null hypothesis is accepted.

→ Head = 10 times (lies outside C.I)
Null hypothesis is rejected.

C.I = [20 - 80]
times head — times head

e.g Whether a person is criminal or not?

① Null hypothesis: A person is not criminal.

② Alternate hypothesis: A person is criminal

③ Experiment/Proof: {DNA, weapons, fingerprint, eye, witness}

↓

Judge will decide based on these proof

↓

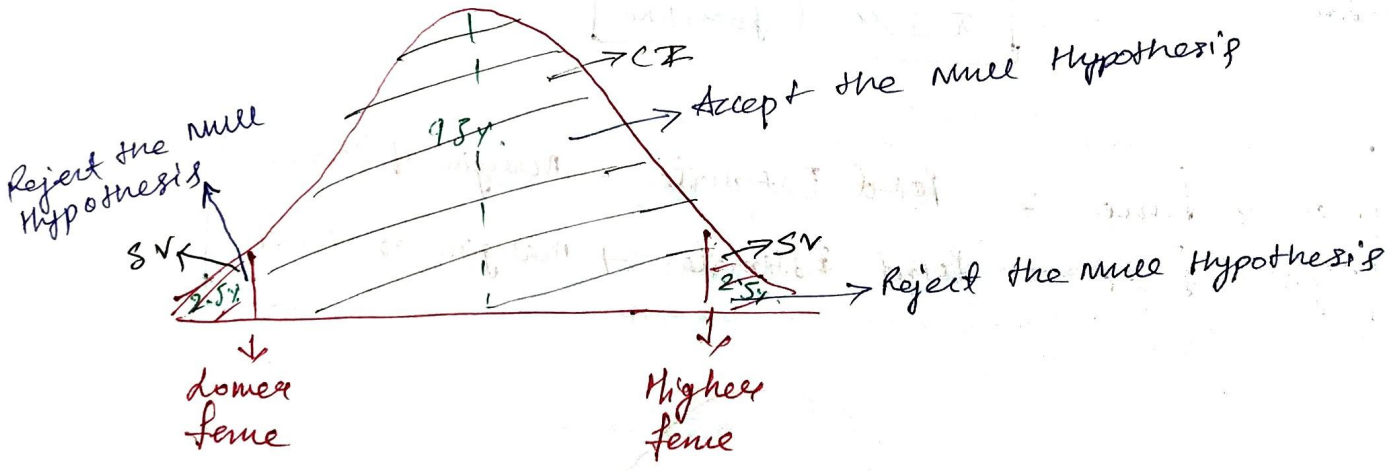
conclusions.

→ When this type of experiment comes with data, we perform test like Z-score, t-test, F-test, chi-square test and Anova test to check.

Confidence Interval (C.I) and Significance Value (SV)

95% CI

Significance Value = 1 - 0.95 = 0.05



→ If value lies b/w lower fence and higher fence, then we will accept null hypothesis else we will reject null hypothesis.

e.g. A examined test needs to be done.

⇒ confidence interval will be very less because medical problems are very very critical.

⇒ confidence interval will be decided by domain expert.

Point Estimate

- The value of any statistics that estimates the value of a parameter is called point estimate.

Sample mean ($\bar{x}$) → Population mean (μ)

$\bar{x}$ —estimating→ μ

Statistics Parameter

→ Based on $\bar{x}$, we are estimating μ.

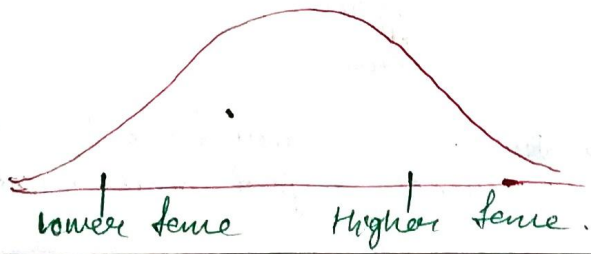
→ Point estimate is helpful in calculating lower fence and higher fence.

Point Estimate ± Margin of Error = Parameter → μ Population mean

$\bar{x}$ Sample mean.

$\bar{x} \geq \mu$ } Both are possible.
$\bar{x} \leq \mu$ }

Lower fence = Point Estimate − Margin of Error
Higher fence = Point Estimate + Margin of Error



$$\text{Margin of Error} = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

α = Significance value = $1 - CI$

$\frac{\sigma}{\sqrt{n}}$ = Standard Error

Population standard deviation give → use Z-test

Qne In the quant test of CAT exam a sample of 25 test takers has a mean of 520 with a ~~sample~~ population standard deviation of ~~80~~ 100. Construct a 95% confidence interval about the mean?

Ans

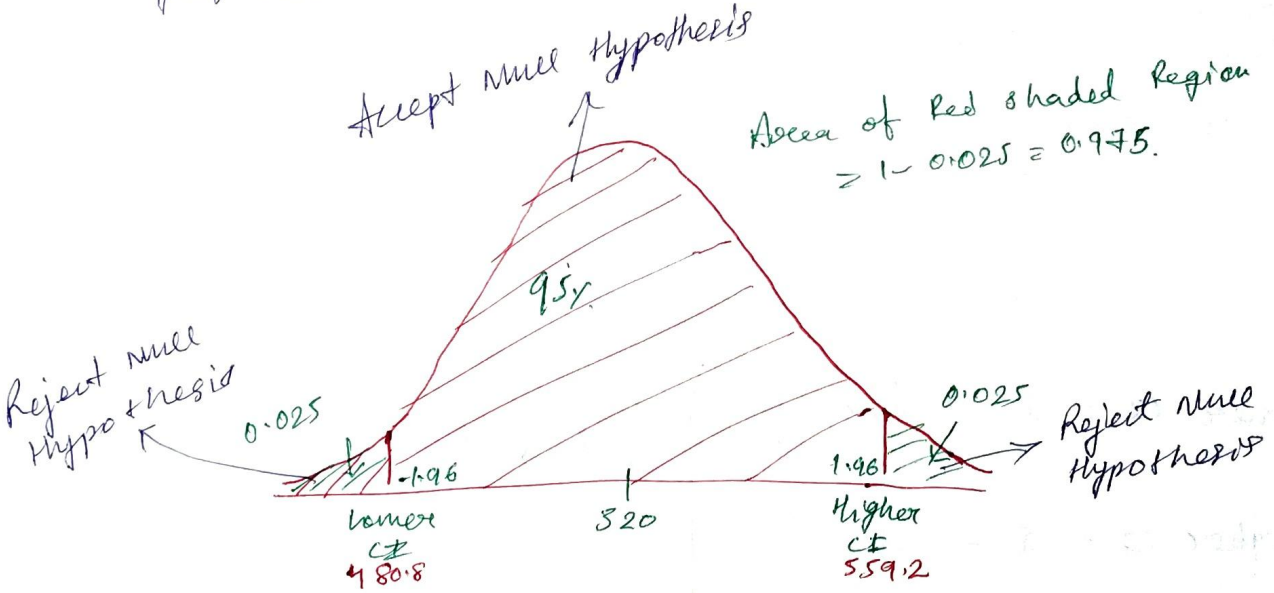
$n = 25$

$\bar{x} = 520$

$\sigma = 100$

$CI = 95\%$

Significance value $= 1 - 0.95 = 0.05$



Lower CI = Point estimate − Margin of Error

$= 520 - Z_{\frac{0.05}{2}} \frac{100}{\sqrt{25}}$

$= 520 - Z_{0.025} \times 20$

$= 520 - 1.96 \times 20$

$= 480.8$

From Z-table,

Area $Z_{0.025} = 1 - 0.025$

Area $= 0.975$

Area $= 0.975$, $Z_{0.025} = 1.96$

Higher CI $= 520 + (+1.96) \times 20$

$= 559.2$

Area $= 0.025$

$Z_{0.025} = -1.96$

→ Within lower CI and higher CI, all the values will be accepted.

Sne $\bar{x} = 480,\ \sigma = 85,\ n = 25,\ CI = 0.90$

Ans Significance value, $\alpha = 1 - 0.90 = 0.10$

$$\text{Lower CI} = 480 - Z_{\frac{0.1}{2}} \frac{85}{\sqrt{25}}$$

$$= 480 - Z_{0.05} \times 17$$

$$= 480 - 1.64 \times 17$$

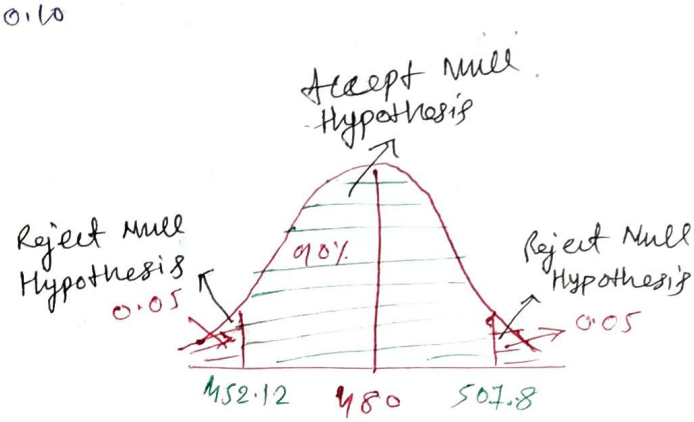
$$= 452.12$$

$$\text{Higher CI} = 480 + Z_{\frac{0.1}{2}} \frac{85}{\sqrt{25}}$$

$$= 480 + Z_{0.05} \times 17$$

$$= 480 + 1.64 \times 17$$

$$= 507.8$$



Area = 0.05
$Z_{0.05}$ = −1.64

Area = 1 − 0.05 = 0.95
$Z_{0.95}$ = 1.64

→ Accept all the values b/w lower CI and Higher CI.

Sne On the quant test of CAT exam, a sample of 25 test takers has a mean of 520, with a sample standard deviation of 80. Construct 95% CI about the mean?

Ans $\bar{x} = 520,\ s = 80,\ CI = 95\%,\ \alpha = 1 - 0.95 = 0.05$

↓ Sample standard Deviation.

↓ We will use t-table.

$$\text{Lower CI} = \bar{x} - t_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

$$\text{Higher CI} = \bar{x} + t_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

→ t-test

Degree of freedom = n − 1 → How many different choices. → meaning of DOF.

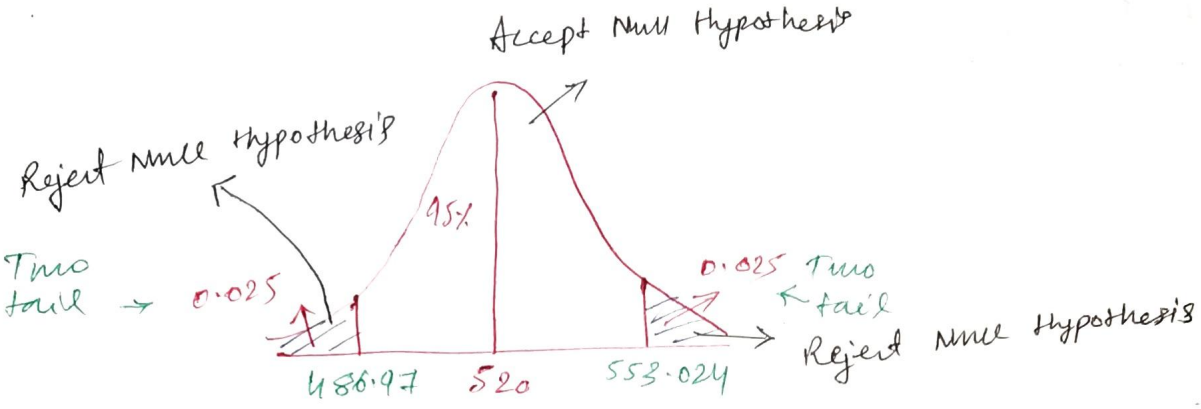
DOF = n − 1 = 25 − 1 = 24

$$\text{Lower CI} = 520 - t_{\frac{0.05}{2}}\left(\frac{80}{\sqrt{25}}\right) = 520 - 2.064 \times 16 = 486.97$$

$$\text{Higher CI} = 520 + t_{\frac{0.05}{2}}\left(\frac{80}{\sqrt{25}}\right) = 520 + 2.064 \times 16 = 553.024$$

for DOF = 24, two tail = 0.05, one tail = 0.025

$t_{0.025} = 2.064$ (take absolute value only) because +/− is already taken into consideration.



NOTE:

- Sample standard deviation given → use t-test
- Population standard deviation given → use z-test

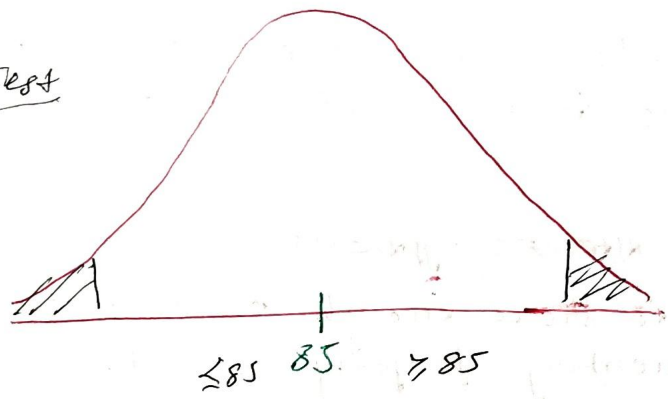
1 Tail and 2 Tail Test

Eg: Colleges in Town A has 85% placement rate. A new college near recently opened and it was found that a sample of 150 students had a placement rate of 88% with a standard deviation of 4%. Does this college has a **different placement** rate with 95% CI?

↓ can be greater than or less than

Ans

Two Tail Test

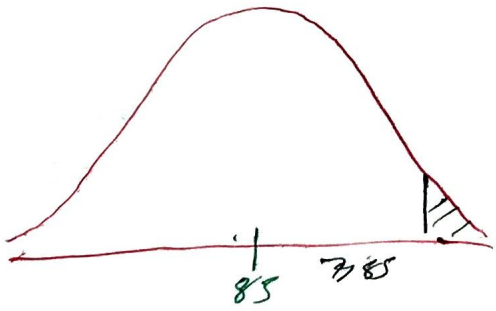


Else Does this college has a **greater than 85%** placement rate with 95% CI ?

↓ only greater

Ans

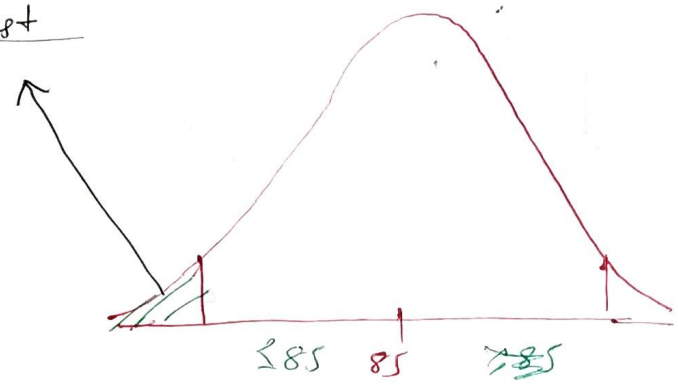
One Tail Test



Que Does this college have a lesser than 85% placement rate with 95% CI?
only lesser

Ans

One Tail Test



Hypothesis Testing

Que A factory has a machine that fills 80 ml of baby medicines in a bottle. An employee believes the average amount of baby medicine is not 80 ml.
Two tail test can be greater than 80 or lesser than 80.

Using 40 sample, he measures the average amount dispersed by the machine to be 78 ml with a standard deviation of 2.5.

(a) State Null and Alternate Hypothesis.

(b) At 95% CI, is there enough evidence to support machine is working properly or not.

$\mu = 80$ ml, $n = 40$, $\bar{x} = 78$ ml, $s = 2.5$

Ans Step 1:

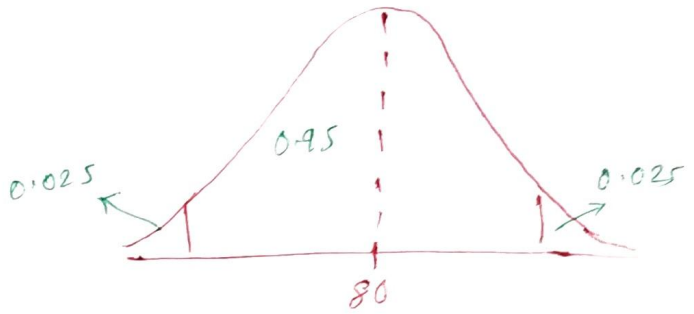
(a) Null Hypothesis : $\mu = 80$ (Machine working properly)

Alternate Hypothesis : $\mu \neq 80$ (Machine not working properly)

Step 2:

$C.I = 0.95$

$\alpha = 1 - 0.95 = 0.05$



Step 3:

$n = 40$

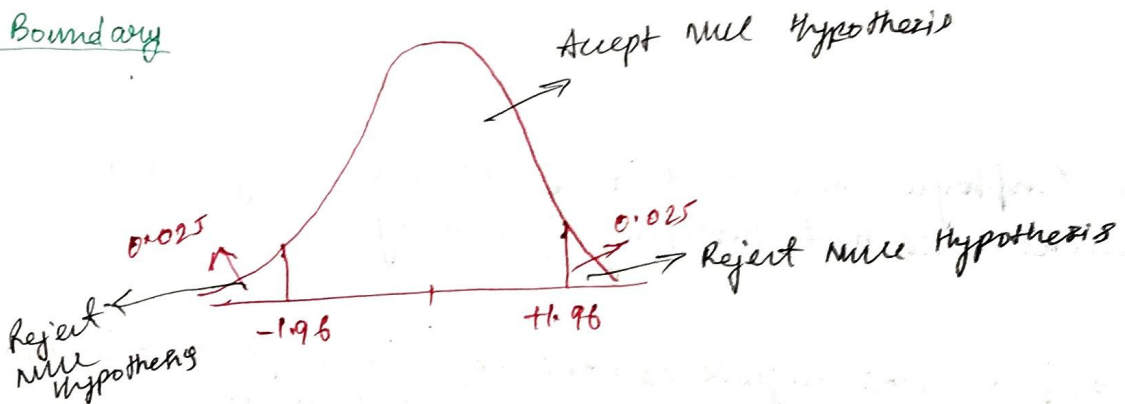
$S = 2.5$ (Sample standard deviation)

[if $n \geq 30$ ~~and~~ or population standard deviation → use z-test

if $n < 30$ and Sample standard deviation → use t-test]

Here, we will use z-test as $n \geq 40$.

Decision Boundary



using z-table,

Area = $1 - 0.025 = 0.975$

$z = 1.96$, for Area = 0.975

Calculate test statistics (z-test)

$$z = \frac{\bar{x} - \mu}{\boxed{\frac{S}{\sqrt{n}}} \rightarrow \text{Standard Error}}$$

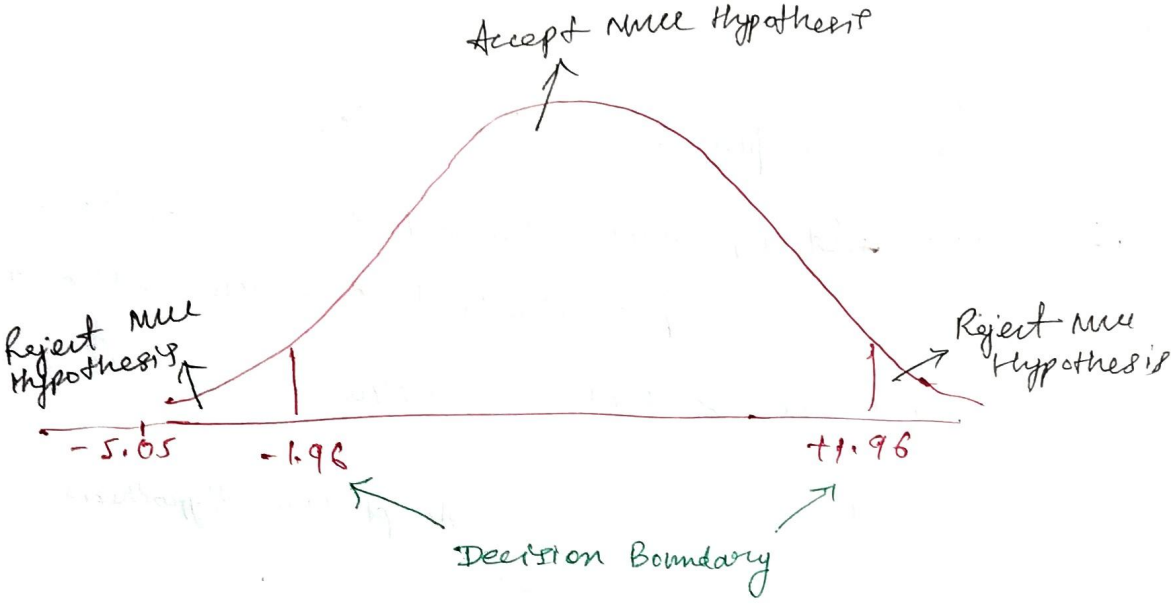
$$= \frac{78 - 80}{\frac{2.5}{\sqrt{40}}}$$

$$z = -5.05$$

Conclusions

Decision Rule: If $Z = -5.05$ is less than -1.96 or greater than $+1.96$, Reject the Null Hypothesis with 95% CI.

→ Reject the Null hypothesis. There is some fault in the machine.



Hence, Employee has said the correct thing, that the machine is not working properly.

Sno A complain was registered, the boys in a government school are underfed. Average weight of the boys of age 10 is 32 kgs with standard deviation of 9 kgs. A sample of 25 boys was selected from the government school. and the average weight was found to be 29.5 kgs?

CI = 95%, check if it is true or false?

Two tail test → check $29.5 > 32$ or $29.5 \leq 2$.

Ans Note!

For Z-test: $n \geq 30$ or Population standard deviation given

For t-test: $n < 30$ and Sample standard deviation given.

Given:

Population standard deviation, $\sigma = 9$

$n = 25$

$\mu = 32$

$\bar{x} = 29.5$.

Step1:

Null Hypothesis, H_0 : $\mu = 32$

Alternate Hypothesis, $H_1 = \mu \neq 32$

Step2:

C.I = 0.95

S.V, $\alpha = 1 - 0.95 = 0.05$

Step3:

We will use z-test.

Two tail test



for Area = 1 - 0.025 = 0.975, Z-score = 1.96 (from z-table)

$$\text{Z-score} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

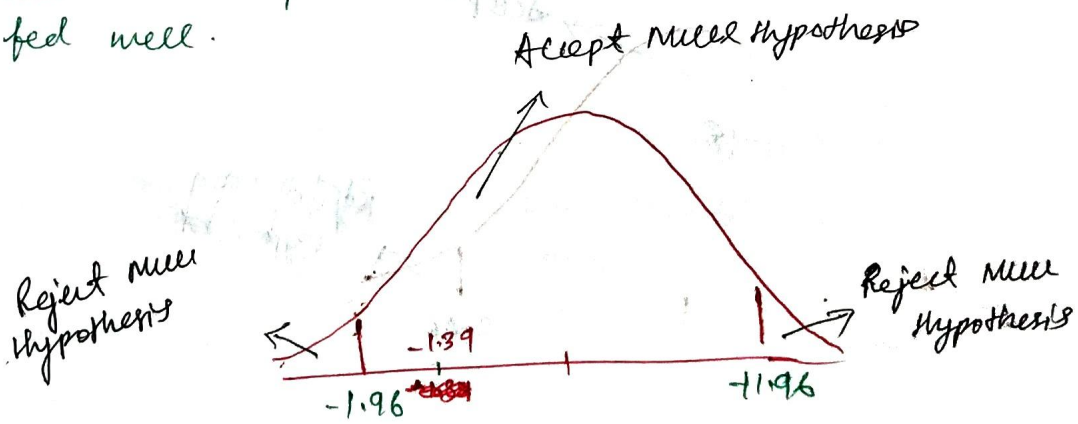
$$= \frac{29.5 - 32}{\frac{9}{\sqrt{25}}}$$

$$= -1.39$$

Conclusion

$-1.39 > -1.96$

We will accept the Null Hypothesis. It means the boys are fed well.



Qne A factory manufactures cars with a warranty of 5 years → one tail or more on the engine and transmission. An engineer test believes that the engine or transmission will malfunction in less than 5 years. He tests a sample of 40 cars → Left one tail test and finds the average time to be 4.8 years with a standard deviation of 0.50.

(a) State the null hypothesis and alternate hypothesis

(b) At a 2% significance level, is there enough evidence to support the idea that the warranty should be revised?

Ans Given:

$n = 40$

$\bar{x} = 4.8$

$\mu = 5$

$s = 0.50$

Step 1:

$H_0: \mu \geq 5$

$H_1: \mu < 5$

Step 2:

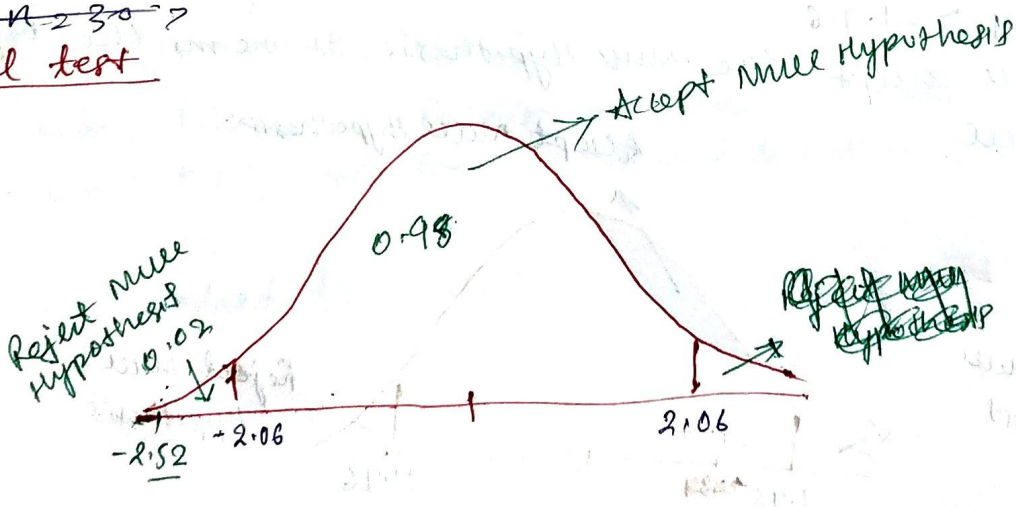
$\alpha = 0.02$

C.I $= 1 - 0.02 = 0.98$

Step 3: ~~$n > 0$~~ $n = 40 > 30$ (we will use Z-test)

~~As $n = 30$?~~

Left-One tail test



From z-table,

Area = the 0.02 & 0.98

z-score z = -2.05 (for area = ~~0.98~~ 0.02)

Calculation

$$z\text{-score } z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{4.8 - 5}{\frac{0.50}{\sqrt{40}}}$$

$$= -2.52$$

Conclusion

$-2.52 < -2.05$

→ we will accept null hypothesis. the warrant should be revoked.

One In the population the average IQ is 100 with a standard deviation of 15. A team of scientists wants to test a new medication to see if it has a (+ve) or (-ve) effect, or no effect at all. A sample of 30 participants who have taken the medication has a mean of 140. Did the medication affect intelligence? C.I = 95%?

↓
Medication affect or not affect intelligence → Two tail test.

Ans Given: $\mu = 100$, $\sigma = 15$, $n = 30$, $\bar{x} = 140$

C.I = 95%

Step 1:

H_0 = Medication affect intelligence

H_1 = Medication doesn't affect intelligence

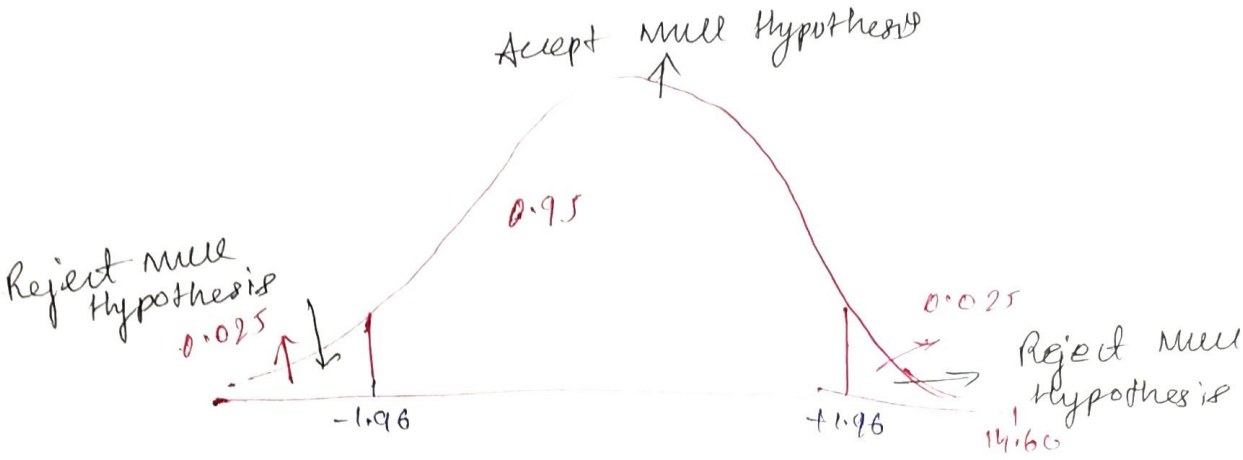
Step 2:

C.I = 0.95

S.V, $\alpha = 1 - 0.95 = 0.05$

Step 3: $n = 30$ and Population standard deviation given we will use z-test.

Two tail test



from z-table,

Area = 1 - ~~0.975~~ 0.025 = 0.975

z-score = 1.96

Calculation

$$z\text{-score} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{140 - 100}{\frac{15}{\sqrt{50}}}$$

$$= 14.60$$

Conclusion

As $14.60 > 1.96$

we will reject null hypothesis and accept new hypothesis. Medication doesn't affect intelligence.

~~Using Z-table~~